

Microstructure Characterization of $\text{Al}_x\text{CoCrCuFeNi}$ High-Entropy Alloy System with Multiprincipal Elements

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A new approach for the design of alloy systems with multiprincipal elements is presented in this research. The $\text{Al}_x\text{CoCrCuFeNi}$ alloys with different aluminum contents (*i.e.*, x values in molar ratio, $x = 0$ to 3.0) were synthesized using a well-developed arc-melting and casting method. These alloys possessed simple fcc/bcc structures, and their phase diagram was predicted by microstructure characterization and differential thermal analyses. With little aluminum addition, the alloys were composed of a simple fcc solid-solution structure. As the aluminum content reached $x = 0.8$, a bcc structure appeared and constructed with mixed fcc and bcc eutectic phases. Spinodal decomposition occurred further on when the aluminum contents were higher than $x = 1.0$, leading to the formation of modulated plate structures. A single ordered bcc structure was obtained for aluminum contents larger than $x = 2.8$. The effects of high mixing entropy and sluggish cooperative diffusion enhance the formation of simple solid-solution phases and submicronic structures with nanoprecipitates in the alloys with multiprincipal elements rather than intermetallic compounds.

I. INTRODUCTION

FOR thousands of years, the development of practical alloy systems has been based mainly on one principal element as the matrix, such as iron-, copper-, and aluminum-based alloys, limiting the number of applicable alloy systems.^[1,2] Even though a substantial amount of other elements is incorporated for property/processing enhancement, superalloys with higher elevated-temperature strengths and better thermal resistance, widely developed since the 1930s, are still based on one principal element such as nickel.^[3] Since the 1970s, intermetallic compounds of Ti-Al, Ni-Al, and Fe-Al binary systems have attracted much attention because of their extremely high specific strengths and thermal resistance.^[4] However, these new compounds with enhanced performance are just based on two principal metallic elements. In the last two decades, many researchers have explored a wide range of bulk amorphous alloys, including Pd-, Ln-, Zr-, Fe-, Mg-based, etc. alloys.^[5–13] The design concept of multicomponent bulk amorphous alloys was, once again, based on one principal element. Some other metallic glasses with multicomponents have been prepared by melt spinning; however, they are still based on one principal group of transition metals.^[14–17]

The main reason for not incorporating multiprincipal elements into alloy preparation is the anticipated formation of

many intermetallic compounds and complex microstructures.^[18] Brittleness of the alloys and also difficulty in processing and analysis are expected with the compound formation and complex microstructures, which discourage other new alloy designs with multiprincipal elements. However, solid solutions with multiprincipal elements will tend to be more stable at elevated temperatures because of their large mixing entropies. Following Boltzmann's hypothesis on the relationship between the entropy and system complexity, the change in configurational entropy during the formation of a solid solution from three elements with an equimolar ratio is already larger than the entropy changes for fusion of most metals.^[19] Consequently, alloys containing a higher number of principal elements will more easily yield the formation of random solid solutions during solidification, rather than intermetallic compounds or other complicated phases.^[20–24]

The promising properties of these types of high-entropy alloys with multiprincipal elements offer the potential to many applications, such as tools, molds, dies, mechanical parts, and furnace parts, which require high strength, thermal stability, and wear and oxidation resistance, with application temperatures up to 800 °C.^[21–25] The alloys can also be used as anticorrosive high-strength materials in chemical plants, IC foundries, and even marine applications for piping components.^[23,25] In addition, coating technology will further expand the application of the alloys to functional films, such as hard facing of golf heads and rollers, diffusion barriers, and soft magnetic films.^[21,23] Therefore, this present study is focused on a new approach to alloy design with multiprincipal elements in equimolar or near-equimolar ratios. Six common metallic elements, including Al, Co, Cr, Cu, Fe, and Ni, were arbitrarily selected. The new $\text{Al}_x\text{CoCrCuFeNi}$ alloys (x : aluminum contents in molar ratio, from 0 to 3.0) were synthesized by conventional arc melting and casting method. Their microstructures and phase transition were examined, and their phase diagram with different aluminum contents was predicted from the microstructure characterization and phase transition temperatures.

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Experimental results have shown us that these alloys exhibit very simple crystal structures and possess submicronic structures with nanoprecipitates under the special effect of high mixing entropy and sluggish cooperative diffusion.

II. EXPERIMENTAL PROCEDURES

The multiprincipal-element $\text{Al}_x\text{CoCrCuFeNi}$ alloy system with different aluminum contents (*i.e.*, x values in molar ratio, from 0 to 3.0) was prepared in this study by arc melting the constituent elements with a current of 500 Amp in a cold copper hearth. Melting and casting were performed in a vacuum of 0.01 atm after purging with argon (Ar) three times. Repeated melting for at least 5 times was carried out to improve chemical homogeneity of the alloys. The alloys were then directly solidified in the cold copper hearth with a cooling rate similar to conventional casting of about 1 to 10 K/s. The solidified ingots were about 50 mm in diameter and 20-mm thick. The weight losses of the alloys after arc melting and casting were lower than 0.2 pct because no volatile element was used for the alloy preparation. These cast ingots were polished and etched with *aqua regia* for observation under an optical microscope and a scanning electron microscope (SEM, JEOL* JSM-5410). The chemical

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compositions of these cast alloys were analyzed by SEM energy dispersive spectrometry. Thin-foil specimens were prepared by mechanical thinning followed by ion milling at room temperature, and subsequently were observed under a transmission electron microscope (TEM, JEOL JEM-2010, Tokyo, Japan) with selected area diffraction (SAD) analysis. An X-ray diffractometer (XRD, Rigaku ME510-FM2) was used for identification of the crystalline structure with the 2θ scan ranging from 20 to 100 deg at a speed of 1 deg/min. The typical radiation condition was 30 kV and 20 mA with a copper target. A differential thermal analyzer (DTA, Perkin-Elmer Instrument, Pyris Diamond, Boston, MA) was used to determine the phase transition temperatures of these alloys scanning from room temperature to 1400 °C at a heating rate of 10 °C/min. Besides using alumina crucibles and providing a sufficient Ar flow of 400 sccm for the protection of the alloys from oxidation, each alloy was prepared as a one-piece sample of about 130 mg, which was large enough to obtain a superior resolution.

III. RESULTS AND DISCUSSION

A. Microstructures of $\text{Al}_x\text{CoCrCuFeNi}$ Alloy System

Figures 1 through 3 show the microstructures of as-cast $\text{Al}_x\text{CoCrCuFeNi}$ alloys with different aluminum contents. Typical cast dendrite and interdendrite structures (defined as DR and ID in the figures, respectively) were observed in these $\text{Al}_x\text{CoCrCuFeNi}$ alloys. The chemical compositions of the cast alloys are listed in Table I, rather close to the designed compositions. Although copper segregation at the interdendrite regions of small volume fractions was observed, the dendrites of large volume fractions were basically composed of multiprincipal elements. With low aluminum contents (x from 0 to 0.5), both the dendrites and interdendrites

appeared to consist of only one simple phase, respectively. As the aluminum content increased to $x = 0.8$, the interdendrites began to decompose into two phases under a eutectic reaction in which a spinodally decomposed netlike structure was observed. After the x values reached 1.0, the dendrites subsequently transformed to clearer two-phase, netlike structures resulting from spinodal decomposition (defined as SD), which are further revealed subsequently by TEM investigation and have been reported by Soffa and Laughlin.^[26] The appearance of the two-phase structure resulted in the drastic change in the morphologies and contrasts of SEM micrographs of the $\text{Al}_{0.8}\text{CoCrCuFeNi}$ and $\text{Al}_{1.0}\text{CoCrCuFeNi}$ alloys shown in Figures 1(d) and (e), which revealed simple-phase dendrites and spinodally decomposed dendrites, respectively. With more addition of aluminum (*i.e.*, increased x values), the amount of the eutectic interdendrites gradually decreased, except a relatively small amount of copper segregation, and the dendrites were mainly constructed of the spinodally decomposed netlike structures, as shown in Figures 2 and 3. Vein structures in polycrystalline structure (defined as Poly) were subsequently observed when the aluminum content exceeded $x = 2.3$, as presented in Figure 3.^[26] More specific characterizations of the microstructures are discussed subsequently in TEM analyses.

From the XRD analyses shown in Figure 4, the crystal structures of the cast $\text{Al}_x\text{CoCrCuFeNi}$ alloy system were characterized, and only simple solid-solution structures, essentially fcc and bcc, were identified. Normally, the heights of fcc (111) peaks were higher than those of fcc (200), but their relative height might be affected by the preferred orientations of the dendritic phases in the alloy samples. It is obviously seen that the bcc phase began to appear at $x = 0.8$, and the ordered bcc phase (B2) began at $x = 1.0$, in consistence with the morphology variation shown in Figures 1 through 3. Although XRD has a detection limit of approximately 1 pct in detecting minor phases,^[27] such as precipitates or inclusions of intermetallic compounds, undoubtedly, almost all alloyed elements were incorporated in the simple solid-solution phases. Particularly, all XRD peak intensities of these alloys were markedly lower than the corresponding ones of conventional alloys under the same XRD measurement conditions, similar to the results of temperature diffused scattering.^[28] Random occupation of variously sized atoms on lattice points caused serious distortion of crystal lattices, and severe X-ray scattering thus occurred on the roughened Bragg diffraction planes, weakening the diffraction signals.^[28,29] Just like the thermal deviation of atoms from their neutral positions, lattice distortion reduced the perfection of the crystalline structure and enhanced scattering, lowering the heights of the diffraction peaks especially those of high-index peaks.

As further confirmed by TEM bright-field image and the corresponding SAD pattern shown in Figure 5(a), both the dendrites and interdendrites of cast $\text{Al}_x\text{CoCrCuFeNi}$ alloys with aluminum contents from $x = 0$ to 0.5, presented in Figures 1(a) through (c), were composed of fcc structure. No special feature but only a strain-field-induced modulated structure formed during TEM sample preparation was observed. An ordered fcc phase with some degree of ordering revealed by the SAD pattern and at the sizes of 5 to 10 nm (bright spots) precipitated within the disordered matrix dur-

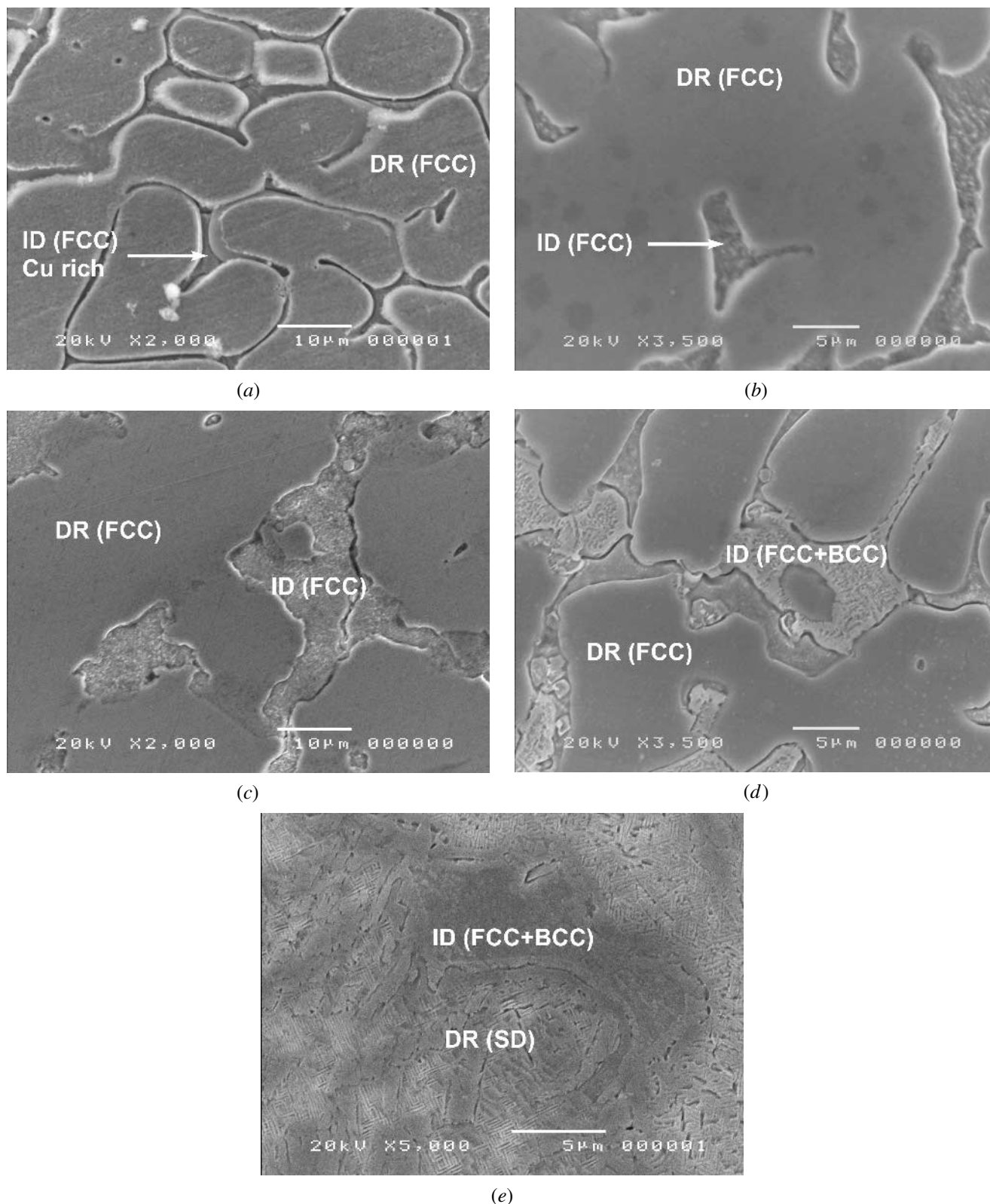


Fig. 1—SEM microstructures of as-cast $\text{Al}_x\text{CoCrCuFeNi}$ alloys with different aluminum contents (x values): (a) 0, (b) 0.3, (c) 0.5, (d) 0.8, and (e) 1.0 (DR: dendrite, ID: interdendrite, and SD: spinodal deposition).

ing solidification, as seen in the TEM dark-field image in Figure 5(b). When the aluminum content of the $\text{Al}_x\text{CoCrCuFeNi}$ alloys reached $x = 0.8$, a bcc structure appeared in addition to the fcc phase, as identified by the XRD analy-

ses presented in Figure 4. Under TEM examination, the dendrites of the $\text{Al}_{0.8}\text{CoCrCuFeNi}$ alloy shown in Figure 1(d) were still characterized to contain only a simple fcc phase, while the extended interdendrites were composed of mixed

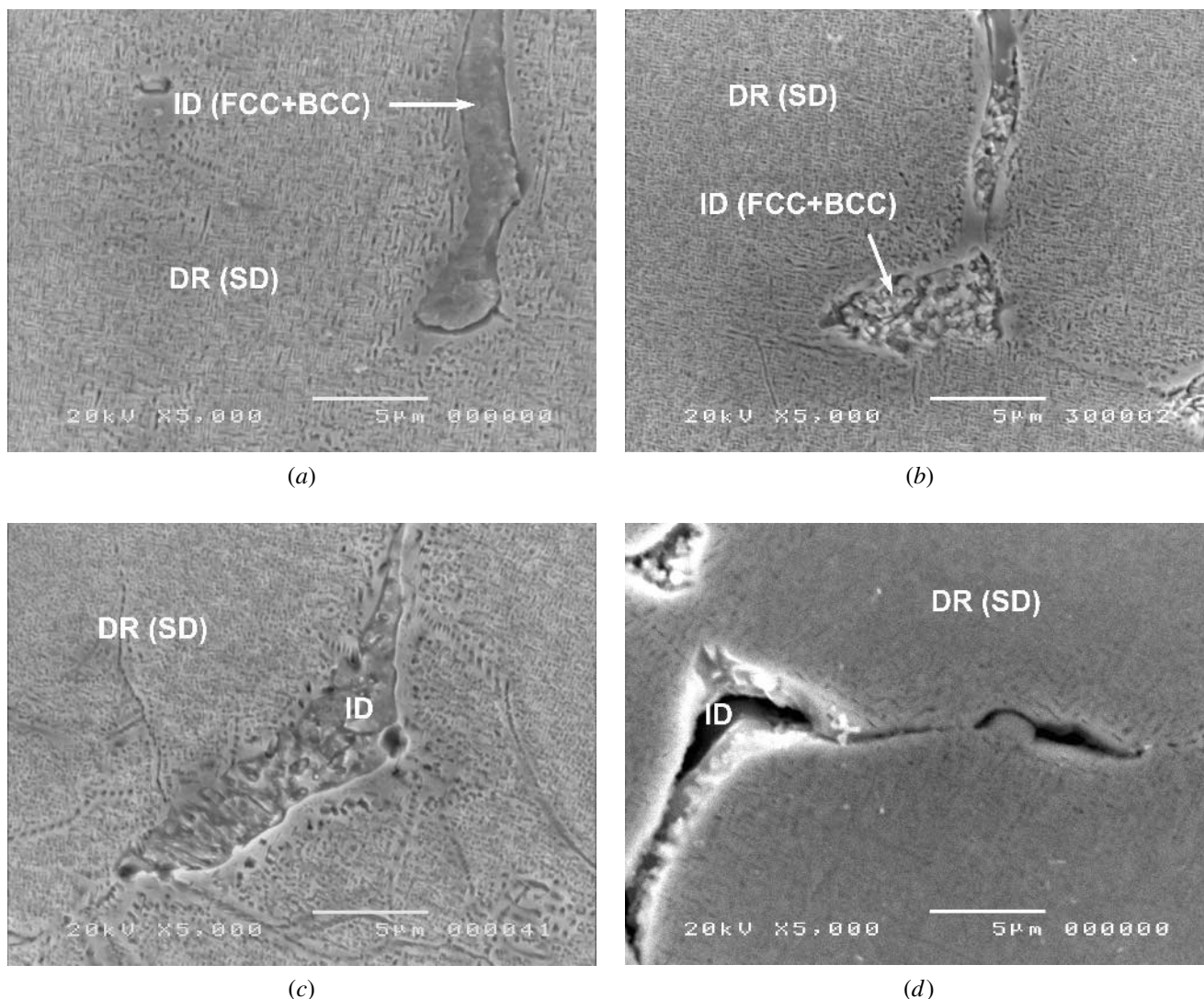


Fig. 2—SEM microstructures of as-cast $\text{Al}_x\text{CoCrCuFeNi}$ alloys with different aluminum contents (x values): (a) 1.3, (b) 1.5, (c) 1.8, and (d) 2.0 (DR: dendrite, ID: interdendrite, and SD: spinodal deposition).

fcc and bcc phases resulting from a eutectic reaction, corresponding to the two main groups of XRD peaks shown in Figure 4. However, it should be mentioned that the bcc phase in the eutectic interdendrites further decomposed into a spinodally decomposed netlike structure. When the aluminum contents (x values) in the $\text{Al}_x\text{CoCrCuFeNi}$ alloys were higher than 1.0, the interdendrites were also composed of the mixed fcc/bcc eutectic phases, whereas spinodal decomposition occurred further on in the bcc dendrites, leading to the formation of modulated plate structures, as presented in the SEM micrograph of the $\text{Al}_{1.0}\text{CoCrCuFeNi}$ alloy in Figure 1(e). By corresponding TEM observation and SAD analyses shown in Figures 6(a) and (b), the modulated plate structures were characterized as being composed of ordered bcc plates (β , 100-nm thick) and disordered bcc interplates (α , 70-nm thick). These ordered bcc plate structures exhibited a preferred orientation of (100). Both the ordered plates and disordered interplates have a bcc structure with the same lattice constants of 2.89 Å but different chemical compositions, confirming the occurrence of the spinodal decomposition of bcc dendrite.^[26,30] Some precipitates found in Figure 6(c)

were verified under SAD pattern analysis close to an fcc structure and, in fact, a face-centered orthorhombic structure ($a = 6.00$ Å, $b = 5.12$ Å, and $c = 7.10$ Å), as seen in Figure 6(d). These precipitates are proposed to be a Cu-rich phase because Cu atoms tend to segregate as clusters during cooling due to their small bonding energies with Fe, Co, Ni, and Cr atoms. Under observation at higher TEM magnification shown in Figure 6(e), the submicronic disordered bcc interplates (α) and ordered bcc plates (β), and nanosized fcc-like precipitates (ϵ , 7 to 50 nm) in the plates, were more clearly identified. The lattice misfit of the nanoprecipitates to the matrix plates was expected to be small so that specific crystallographic coherence existed between them and thus the nanoprecipitates exhibited a round shape.^[30] Some other nanoscaled contrasts (δ) in the interplates were formed due to the surface roughness effect of TEM foil since their density seemed unchanged with the foil thickness.

With a greater addition of aluminum (*i.e.*, increased x values), the amount of fcc/bcc eutectic phases in the $\text{Al}_x\text{CoCrCuFeNi}$ alloy system gradually decreased, and the bcc phase gradually tended to be the main composing phase,

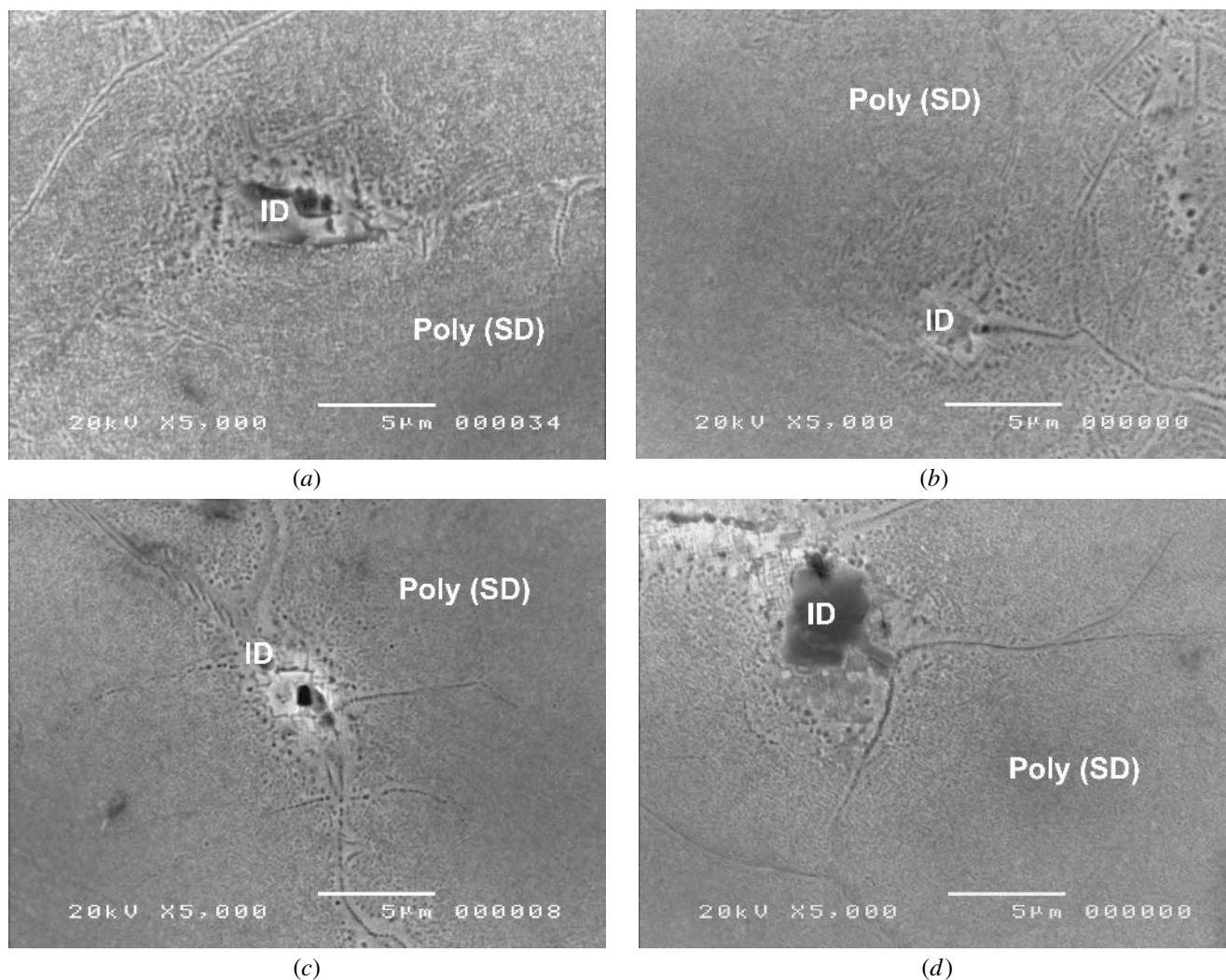


Fig. 3—SEM microstructures of as-cast $\text{Al}_x\text{CoCrCuFeNi}$ alloys with different aluminum contents (x values): (a) 2.3, (b) 2.5, (c) 2.8, and (d) 3.0 (Poly: polycrystalline structure, ID: interdendrite, and SD: spinodal deposition).

Table I. Chemical Compositions of Cast $\text{Al}_x\text{CoCrCuFeNi}$ Alloy System in Atomic Percentage

Alloy	Element (At. Pct)	Al	Co	Cr	Cu	Fe	Ni
$\text{Al}_{0.0}\text{CoCrCuFeNi}$	total	0	20.9	21.4	16.5	20.8	20.5
	dendrite	0	23.1	22.8	10.6	22.9	20.6
	interdendrite	0	3.8	3.2	78.5	4.1	10.4
$\text{Al}_{0.5}\text{CoCrCuFeNi}$	total	9.4	17.7	18.2	17.7	18.0	19.0
	dendrite	9.4	18.7	19.5	12.6	19.4	20.4
	interdendrite	12.5	5.2	4.1	61.3	4.7	12.2
$\text{Al}_{1.0}\text{CoCrCuFeNi}$	total	16.7	17.0	17.2	15.6	16.8	16.8
	dendrite	16.7	17.1	20.7	11.0	18.0	16.5
	interdendrite	12.3	5.2	3.0	62.5	5.1	11.9
$\text{Al}_{1.5}\text{CoCrCuFeNi}$	total	24.3	15.5	15.6	13.7	15.6	15.3
	dendrite	24.3	16.4	16.5	10.6	16.3	15.9
	interdendrite	15.9	4.3	2.6	65.4	3.7	8.1
$\text{Al}_{2.0}\text{CoCrCuFeNi}$	total	28.5	13.8	13.9	13.5	15.5	14.8
	dendrite	29.2	14.0	17.6	10.8	15.3	13.1
	interdendrite	19.1	2.7	1.5	69.2	2.6	4.9
$\text{Al}_{2.5}\text{CoCrCuFeNi}$	total	36.3	13.0	11.8	11.3	14.1	13.4
	dendrite	35.8	14.6	12.1	9.1	14.3	14.0
	interdendrite	19.0	2.3	1.5	71.3	2.7	3.3
$\text{Al}_{3.0}\text{CoCrCuFeNi}$	total	38.9	12.6	11.2	11.7	13.5	12.1
	dendrite	39.1	13.2	12.2	9.2	14.2	12.1
	interdendrite	35.7	6.2	2.6	39.9	7.0	8.6

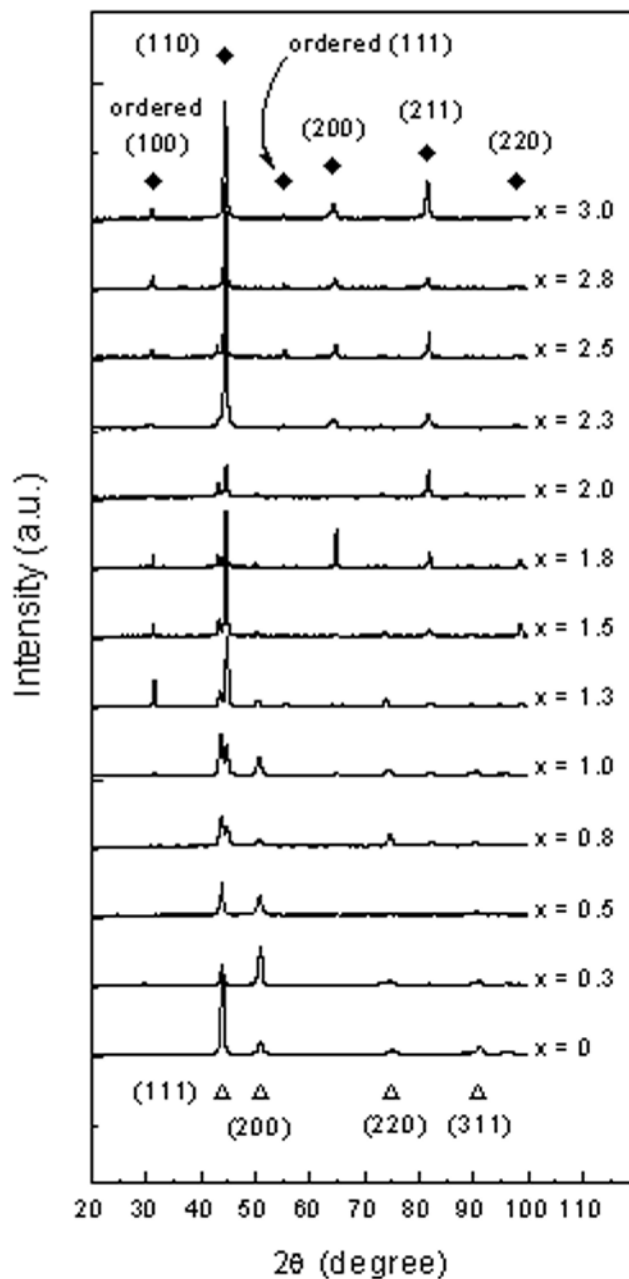
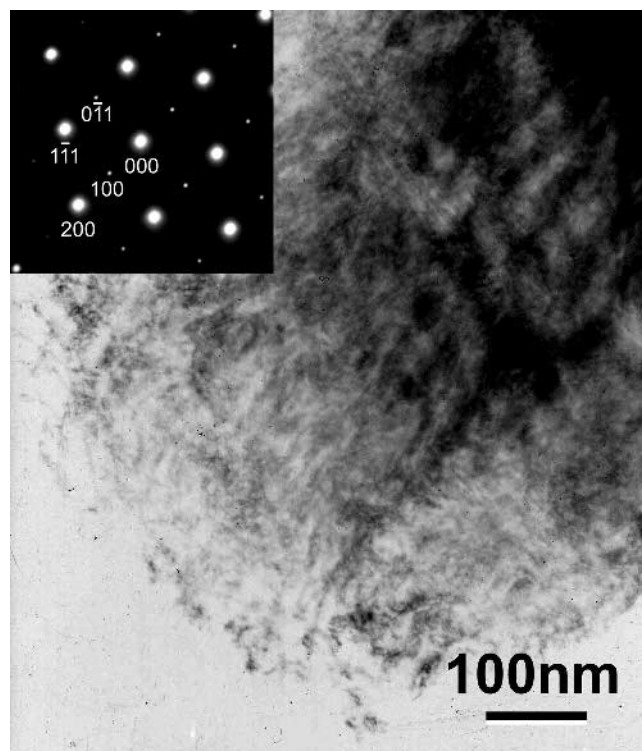
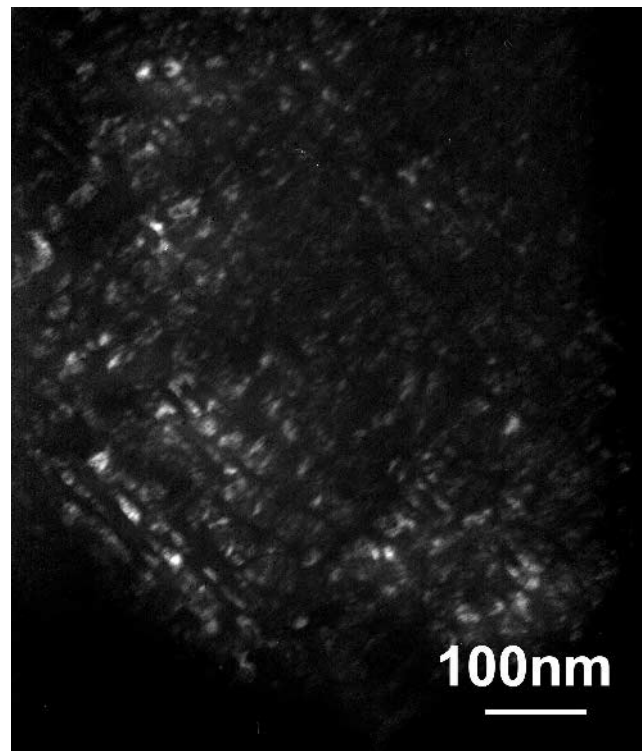


Fig. 4—XRD analyses of $\text{Al}_x\text{CoCrCuFeNi}$ alloy system with different aluminum contents (x values); Δ fcc phase, and $\blacklozenge$ bcc phase (ordered peaks of B2 phase are also indicated); peak intensity between each two minor ticks is 500 cps.

as shown in the XRD patterns in Figure 4. Previously mentioned modulated plate structures resulting from spinodal decomposition in dendrites transformed into a matrix dispersed with long precipitates as shown in the TEM image of the $\text{Al}_{1.5}\text{CoCrCuFeNi}$ alloy in Figure 7(a). It was more clearly found at higher magnification in Figure 7(b) and by SAD analyses in Figures 7(c) and (d) that the ordered bcc plates transformed into the matrix, and the disordered bcc interplates formed long Widmanstätten precipitates (about 30 nm in width) along the direction of $\langle 100 \rangle$. In addition, some other nanosized spherical precipitates (5 to 15 nm) of disordered bcc phase were also found in the ordered matrix. With aluminum content up to $x = 2.0$, as shown in Figure 8,



(a)



(b)

Fig. 5—TEM microstructures of the dendrite of as-cast $\text{Al}_{0.5}\text{CoCrCuFeNi}$ alloy: (a) bright-field image with SAD pattern of fcc [011] zone axis and (b) dark-field image corresponding to the (100) superlattice spot of the SAD pattern shown in (a).

the dendrites in $\text{Al}_{2.0}\text{CoCrCuFeNi}$ alloy were still constructed of spinodally decomposed structures in which disordered Widmanstätten precipitates in the ordered bcc matrix became

thinner (about 15 nm in width). Vein structures presented in Figure 3 subsequently formed by spinodal decomposition during which spherical particles precipitated when the aluminum content exceeded $x = 2.3$, and a single bcc structure with ordered peaks was obtained for aluminum contents larger than $x = 2.8$ from XRD analyses shown in Figure 4. The TEM images of the $\text{Al}_{3.0}\text{CoCrCuFeNi}$ alloy in Figure 9 show no special feature but much stronger ordering of the ordered bcc phase corresponding to (100) superlattice spots. Disordered long Widmanstätten precipitates transformed into a spherical shape due to small elastic anisotropy and dispersed in the strongly ordered matrix as arrowed in Figure 9(b). It is interesting to note that the various morphologies observed with different aluminum contents, $x = 1.0$ to 3.0, were well consistent with the typical morphologies of continuous transformations in alloys reported by Soffa and Laughlin.^[26]

B. Phase Diagram

Conclusively from the results of cast microstructures and XRD analyses, the microstructure characterization of the solidified $\text{Al}_x\text{CoCrCuFeNi}$ alloy system with different aluminum contents (*i.e.*, x values in molar ratio) is summarized in Table II. In combination with the data obtained by differential thermal analyses, including ordering temperatures, spinodal decomposition temperatures, melting points of Cu-rich

phase, and solidus and liquidus temperatures, as listed in Table III, the phase diagram of the present alloy system was constructed as shown in Figure 10. Although the Cu-rich fcc phase existed below the melting points of the alloy system, its amount was relatively small as compared to other phases. With few additions of aluminum ($x = 0$ to 0.5), the structure was mainly composed of disordered fcc phase in which an ordered fcc phase formed during cooling. The ordered fcc phase was suggested as the L1_2 structure, similar to the ordered fcc structure of the Ni_3Al intermetallic compound.^[30] However, even with the aluminum content of $x = 0.5$, the amount of aluminum was still insufficient to form near-stoichiometric Me_3Al (Me: other metallic elements). Thus, some of the other five metallic elements were expected to supplement aluminum to form the observed ordered L1_2 structure as solid-solution configuration.^[31] With greater addition of aluminum atoms (higher x values), the crystal structure of the alloy system gradually transformed from fcc into stabilized bcc. A eutectic point was expected to exist at an aluminum content between $x = 0.8$ and 1.0, and spinodal decomposition then occurred with aluminum contents higher than $x = 1.0$. The morphology of the spinodal decomposition changed with increasing aluminum content (x value), meanwhile, an ordered B2 structure more easily formed. The ordered B2 structure was similar to the ordered bcc structure of NiAl intermetallic compound.^[30] Similarly, even with the aluminum content of $x = 3.0$, some of the other five metallic elements were

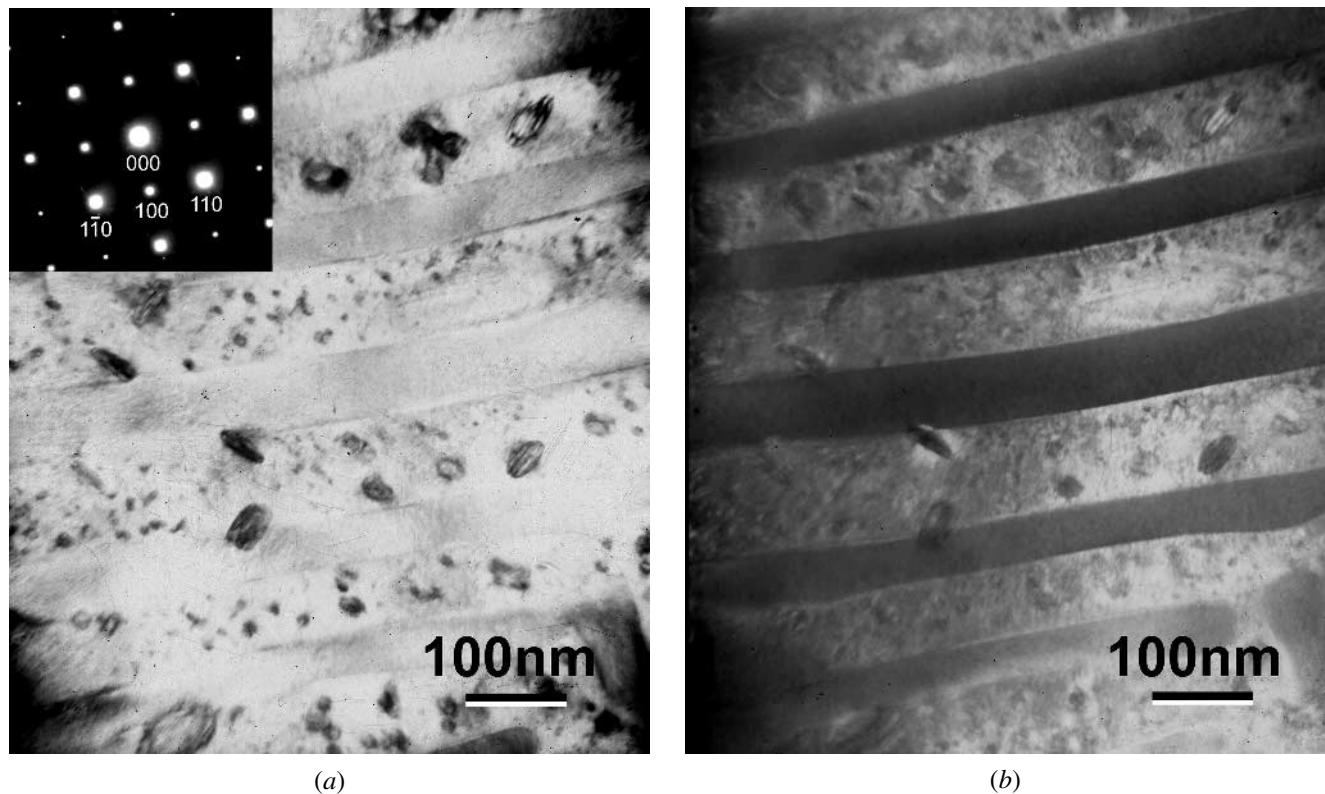
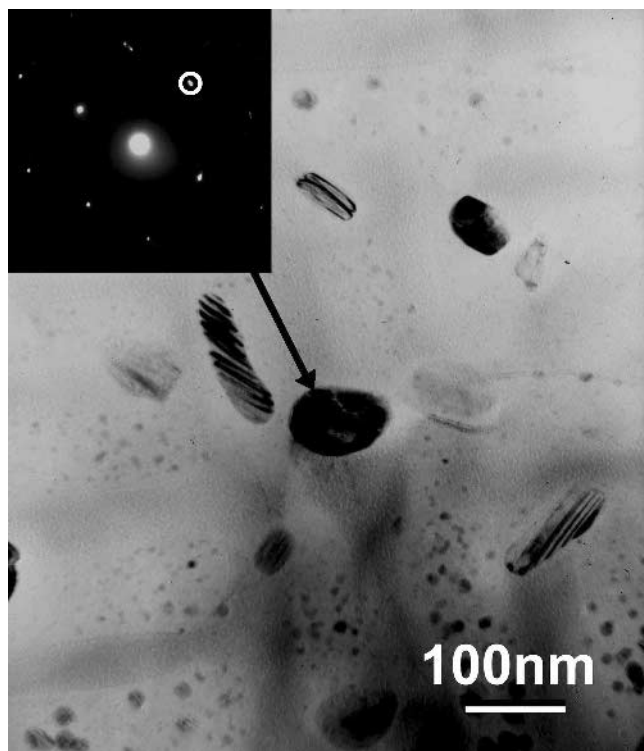
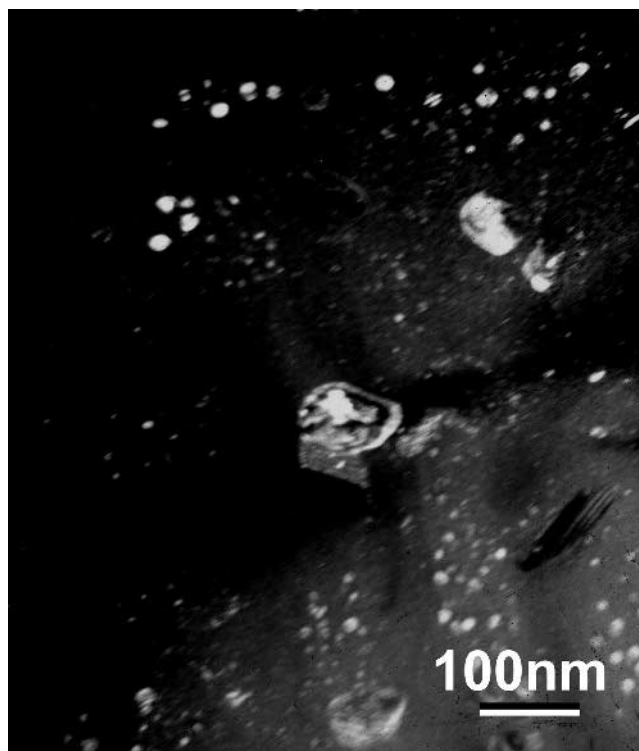


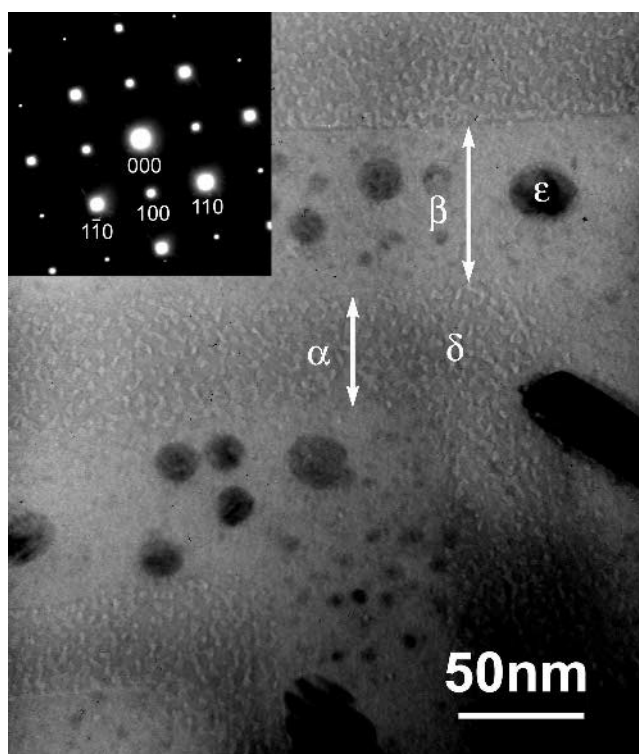
Fig. 6—TEM microstructures of the dendrite of as-cast $\text{Al}_{1.0}\text{CoCrCuFeNi}$ alloy: (a) bright-field image of modulated plates and interplates with SAD pattern of bcc [001] zone axis; (b) dark-field image corresponding to the (010) superlattice spot of the SAD pattern shown in (a); (c) bright-field image with SAD pattern of arrowed precipitate close to fcc $[\bar{2}33]$ zone axis; (d) dark-field image corresponding to the circled spot of the SAD pattern shown in (c); and (e) enlarged bright-field image of the modulated plates and interplates with the same SAD pattern of bcc [001] zone axis shown in (a) (α : interplate, 70-nm wide, disordered bcc phase (A_2), and lattice constant 2.89 Å; β : plate, 100-nm wide, ordered bcc phase (B_2), lattice constant 2.89 Å, ϵ : nanoprecipitation in plate, 7 nm to 50 nm in diameter, close to fcc phase, and δ : contrast in interplate due to surface roughness effect of TEM foil).



(c)

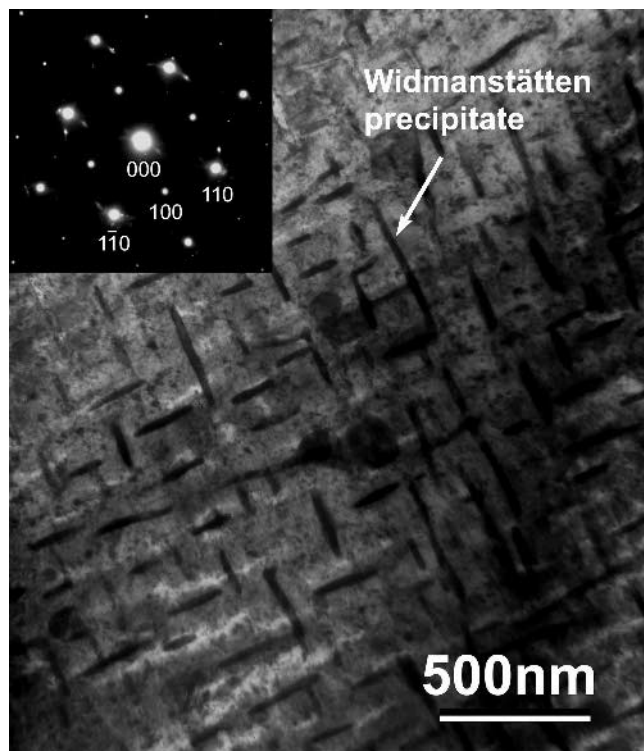


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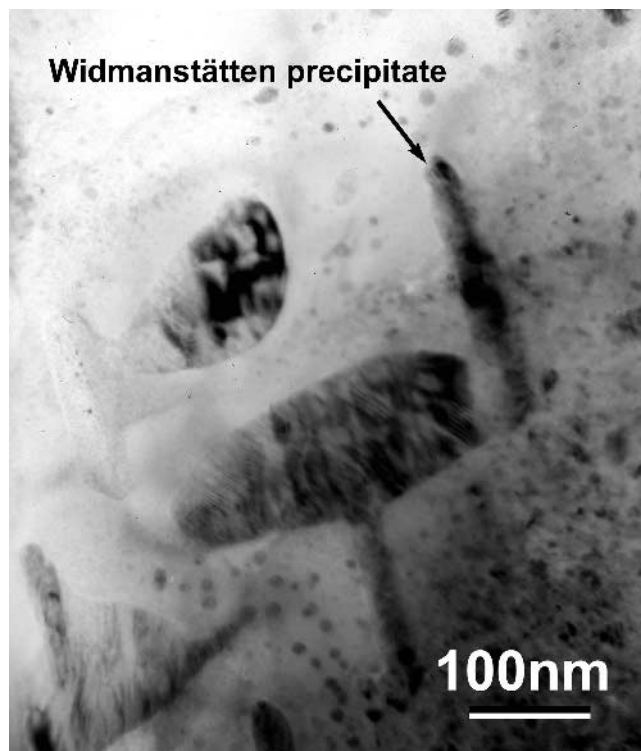


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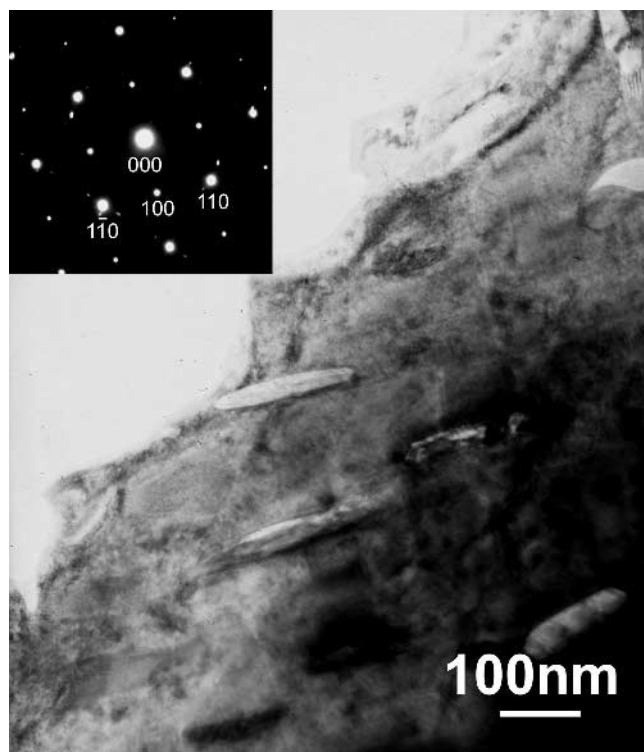
Fig. 6—(Continued). TEM microstructures of the dendrite of as-cast $\text{Al}_{1.0}\text{CoCrCuFeNi}$ alloy: (a) bright-field image of modulated plates and interplates with SAD pattern of bcc [001] zone axis; (b) dark-field image corresponding to the (010) superlattice spot of the SAD pattern shown in (a); (c) bright-field image with SAD pattern of arrowed precipitate close to fcc [233] zone axis; (d) dark-field image corresponding to the circled spot of the SAD pattern shown in (c); and (e) enlarged bright-field image of the modulated plates and interplates with the same SAD pattern of bcc [001] zone axis shown in (a) (α : interplate, 70-nm wide, disordered bcc phase (A_2), and lattice constant 2.89 Å; β : plate, 100-nm wide, ordered bcc phase (B_2), lattice constant 2.89 Å, ϵ : nanoprecipitation in plate, 7 nm to 50 nm in diameter, close to fcc phase, and δ : contrast in interplate due to surface roughness effect of TEM foil).



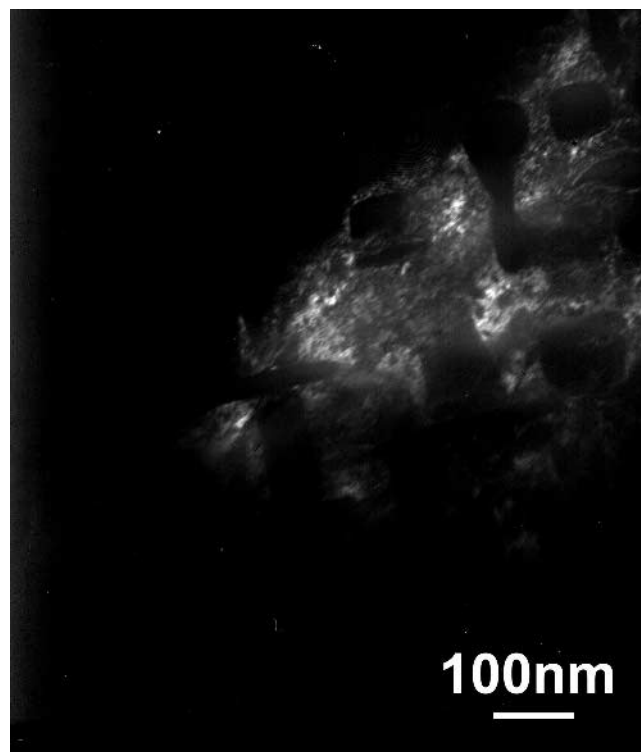
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(b)



(c)

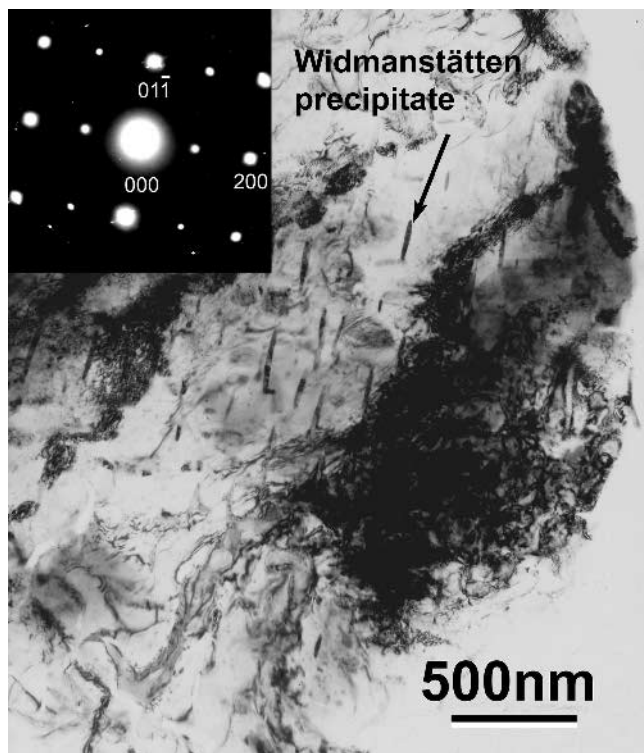


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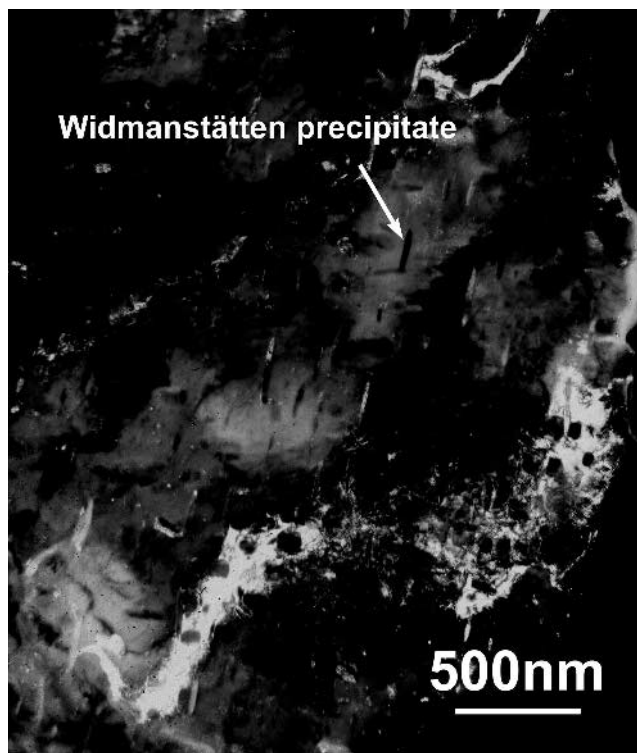
Fig. 7—TEM microstructures of the dendrite of as-cast $\text{Al}_{1.5}\text{CoCrCuFeNi}$ alloy: (a) bright-field image with SAD pattern of bcc [001] zone axis, (b) bright-field image showing disordered bcc Widmanstätten precipitates and nanoprecipitates (5 to 15 nm), (c) bright-field image with SAD pattern of bcc [001] zone axis, and (d) dark-field image corresponding to the (100) superlattice spot of the SAD pattern shown in (c).

required to supplement the insufficient amount of aluminum to form the observed ordered B2 structure as near-stoichiometric MeAl configuration.^[31] It should be noted that the phase

diagram of multiple components only suggests the phases existing in equilibrium at different temperatures, but not their compositions coinciding with the tie-lines by lever rule.^[30]

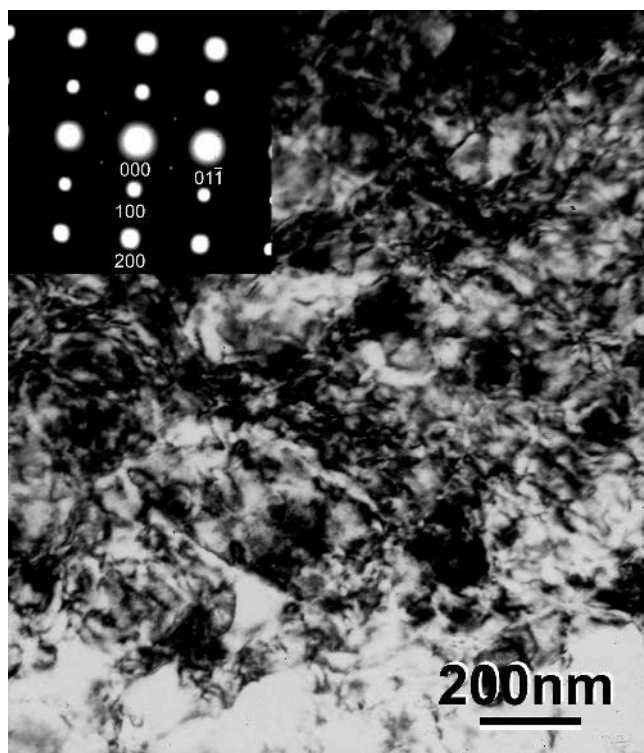


(a)

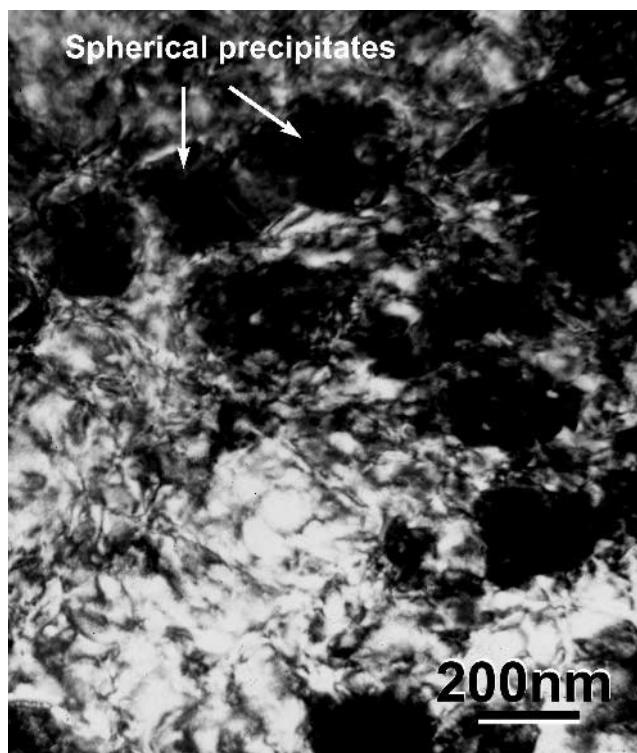


(b)

Fig. 8—TEM microstructures of the dendrite of as-cast $\text{Al}_{2.0}\text{CoCrCuFeNi}$ alloy: (a) bright-field image with SAD pattern of bcc [011] zone axis and (b) dark-field image corresponding to the (100) superlattice spot of the SAD pattern shown in (a).



(a)



(b)

Fig. 9—TEM microstructures of the dendrite of as-cast $\text{Al}_{3.0}\text{CoCrCuFeNi}$ alloy: (a) bright-field image with SAD pattern of bcc [011] zone axis and (b) dark-field image corresponding to the (100) superlattice spot of the SAD pattern shown in (a).

Table II. Microstructure Characterization of $\text{Al}_x\text{CoCrCuFeNi}$ Alloy System

Al Content (x)	0 to 0.5	1.0 to 1.3	1.5 to 2.0	2.3 to 3.0
Crystal structure	fcc	fcc + bcc	fcc + bcc	bcc
Minor phase	fcc (Cu rich)	fcc (Cu rich)	fcc (Cu rich)	fcc (Cu rich)
Spinodal decomposition	no	intervened plate structure	Widmanstätten precipitates	spherical precipitates
Ordered phase	nanoprecipitates	plates	matrix	matrix
Disordered phase	matrix	interplates	Widmanstätten precipitates	spherical precipitates
Preferred orientation	yes	yes	yes	yes

Table III. Phase Transition Temperatures of the $\text{Al}_x\text{CoCrCuFeNi}$ Alloy System Measured by DTA (Temperature Limit of Measurement: 1400 °C)*

Alloy	T_{order} (°C)	T_{sp}^1 (°C)	T_{sp}^2 (°C)	T_m^{Cu} (°C)	T_m^s (°C)	T_m^l (°C)
$\text{Al}_{0.0}\text{CoCrCuFeNi}$	692	—	—	1135	1339	1389
$\text{Al}_{0.5}\text{CoCrCuFeNi}$	695	—	—	1153	1279	1362
$\text{Al}_{1.0}\text{CoCrCuFeNi}$	—	580	660	1117	1170	1330
$\text{Al}_{1.5}\text{CoCrCuFeNi}$	—	577	648	1086	1259	1355
$\text{Al}_{2.0}\text{CoCrCuFeNi}$	—	—	—	1077	1303	>1400
$\text{Al}_{2.5}\text{CoCrCuFeNi}$	—	—	—	1078	1260	>1400
$\text{Al}_{3.0}\text{CoCrCuFeNi}$	—	585	610	1096	1293	>1400

* T_{order} : ordering temperature, T_{sp} : spinodal decomposition temperature, T_m^{Cu} : melting point of Cu-rich phase, T_m^s : solidus temperature, and T_m^l : liquidus temperature

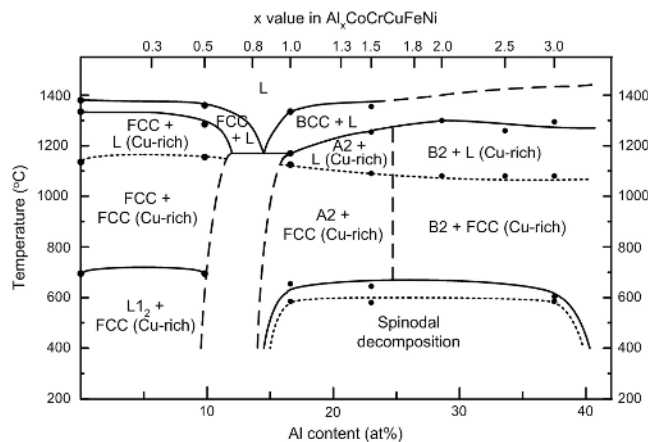


Fig. 10—Predicted phase diagram of $\text{Al}_x\text{CoCrCuFeNi}$ alloy system with different aluminum contents (x values). L: liquid phase. The phase transition temperatures of the alloys were measured by DTA (temperature limit of measurement: 1400 °C) and indicated as black solid dots in the figure.

Although the phase diagram of the $\text{Al}_x\text{CoCrCuFeNi}$ alloy system was constructed based on the solidified microstructures under arc melting and differential thermal analyses, some other more accurate methods would be required to build a more precise phase diagram. However, the present information still tells us the basic structure of the phase diagram with varied temperature and aluminum content (x value). It is also noticed that simple phase regions can exist in such an alloy system with multiprincipal elements. According to the phase rule $F = C - P + 1$ (F : degree of freedom, C : number of components, and P : number of phases),^[19] the freedom of any single phase in this six-component alloy system equals six, and that of the two-phase region equals five. This finding means that, except temperature, the concentrations of the six components in these

solid-solution phases can be adjusted within some specific ranges under which these single phases can still remain. It suggests that the phase regions in Figure 10 could be thus extended within some suitable ranges of other dimensions. This might benefit the study of the effect of solid solutions on various properties.

C. On the Formation of Simple Solid-Solution Structures

Based on the experience with conventional alloys containing one, two, or three major elements, large numbers of intermetallic compounds or terminal solid-solution phases that are solutions with a respective principal matrix element are expected to form in multicomponent alloy systems with more major elements.^[18,32] However, the resultant phases in the $\text{Al}_x\text{CoCrCuFeNi}$ alloy system were relatively simple, as mentioned previously. The superiority of simple solid solutions over intermetallic compounds and terminal solid solutions was attributed to the significant lowering of free energy by the high entropy of mixing.^[19,23,30]

In view of the Hume–Rothery rules for a high degree of solubility between two elements,^[33] this $\text{Al}_x\text{CoCrCuFeNi}$ alloy system has revealed the adequate enhancement on the mutual solubility among multiprincipal elements by high mixing entropy. The Hume–Rothery rules define that, for these solute elements, (1) the atomic size differences must be within 15 pct of each other, (2) the type of crystal structure must be the same, (3) the chemical valence must differ by no more than one, and (4) the electronegativities must be nearly equal.^[33] As listed in Table IV,^[34,35] it is clearly seen that rules (2) and (3) are not fulfilled among many pairs of elements such as Cu–Co, Co–Cr, Ni–Cr, Ni–Fe, and Fe–Al to assure a high mutual solubility. Thus, the fact that these simple solution phases consist of multiprincipal elements in the $\text{Al}_x\text{CoCrCuFeNi}$ alloys indeed demonstrates the effect of high mixing entropy with multiprincipal elements

Table IV. Basic Properties of Elements Used in the $\text{Al}_x\text{CoCrCuFeNi}$ Alloy System^[34,35]

Element	Al	Co	Cr	Cu	Fe	Ni
Atomic size (Å)	1.43	1.25	1.25	1.28	1.24	1.25
Crystal structure at 25 °C	fcc	hcp	bcc	fcc	bcc	fcc
Chemical valence	+3	+2, +3	+3, +6	+1, +2	+2, +3	+2
Electronegativity	1.5	1.8	1.6	1.9	1.8	1.8

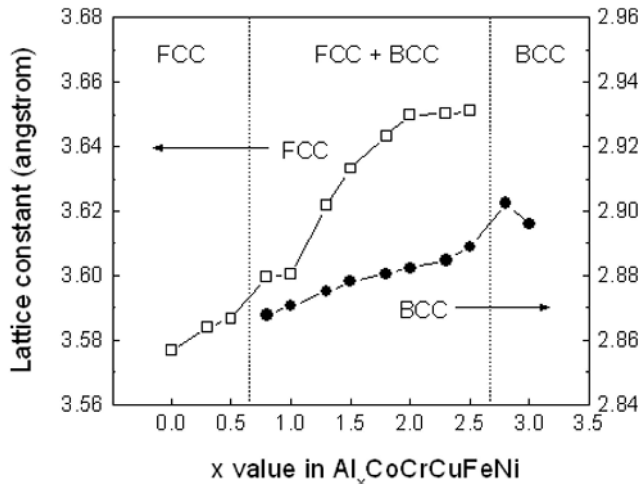


Fig. 11—Lattice constants of $\text{Al}_x\text{CoCrCuFeNi}$ alloy system with different aluminum contents (x values), □ fcc phase and ● bcc phase.

on extending the mutual solubility between elements rather than forming terminal solid solutions. This means the constraints set by the Hume–Rothery rules between two elements seem to be probably relaxed by the effect of high mixing entropy with multiprincipal elements.

Figure 11 shows the lattice constants of fcc and bcc phases in the $\text{Al}_x\text{CoCrCuFeNi}$ alloy system, which were calculated from the measured XRD peak positions by extrapolating the data to $\theta = 90^\circ$.^[28] The lattice constants of both the fcc and bcc phases increased with increasing content of the largest aluminum atom, further clearly confirming the solid-solution phenomenon of aluminum atoms into the alloy structures and thus an expected large extent of lattice strain. This also implies that a high mixing entropy balanced the lattice strain to some extent. Besides, a high-entropy effect will, especially at high temperatures, make solid-solution phases more stable than ordered intermetallic compounds with stoichiometric or near-stoichiometric ratios. The B2 phase presented in this study itself also has an enhanced configurational entropy owing to its simultaneous incorporation of several transition elements. It is generally recognized that an alloy system with a complex microstructure containing a large number of different phases will be difficult to analyze and manipulate.^[18] Thus, the absence of complex phases in this $\text{Al}_x\text{CoCrCuFeNi}$ alloy system indicates the significance of simple-structured alloys incorporated with multiprincipal elements to be further developed.^[20,23]

D. On Precipitation

It has been shown that the present $\text{Al}_x\text{CoCrCuFeNi}$ alloy system possesses ultrafine precipitation in the matrix. Ordered fcc nanoprecipitates disperse in disordered fcc matrix with low

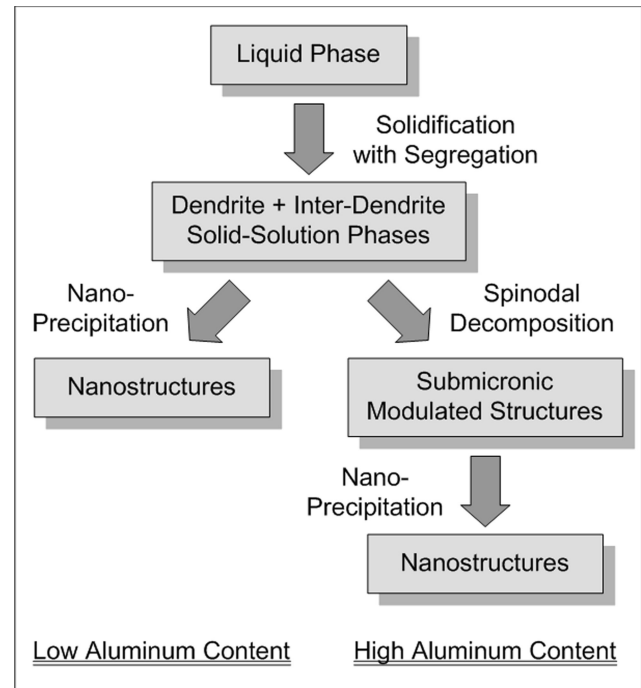


Fig. 12—Depiction of phase formation sequence during cooling of $\text{Al}_x\text{CoCrCuFeNi}$ alloy system with different aluminum contents.

aluminum contents (small x values), and nanosized phases precipitate in fine spinodally decomposed phases with high aluminum contents (x values). The occurrence of precipitation is believed to follow the decreased solubility of the fcc and bcc matrix phases with lowering temperature due to the diminishing effect of high mixing entropy. Figure 12 depicts the phase formation sequence during cooling of the $\text{Al}_x\text{CoCrCuFeNi}$ alloy system. It should be mentioned that, based on diffusion efficiency, the sizes of these precipitates are in nanoscale but not as coarse as usually found in conventional cast alloys. In the matrix containing multiprincipal elements, phase transformation such as spinodal decomposition and precipitation requires cooperative diffusion of many different kinds of atoms to accomplish the partition of composition. The movement of substitutional solute atoms in such diffusion is expected to be especially sluggish so that phase separation proceeds slowly to yield the observed submicronic structures with nanoprecipitates.

IV. CONCLUSIONS

In this research, a new approach for the design of alloy systems with multiprincipal elements was presented. The $\text{Al}_x\text{CoCrCuFeNi}$ alloys with different aluminum contents (*i.e.*, x values in molar ratio) were synthesized using a well-

developed arc-melting and casting method, and their microstructures were characterized. By microstructure characterization and X-ray diffraction analyses, these alloys were found to possess simple fcc/bcc submicronic structures, and in combination with differential thermal analyses, their phase diagram with different aluminum contents was predicted. With aluminum contents less than $x = 0.5$, the alloys were composed of a simple fcc solid-solution structure. As the aluminum content reached $x = 0.8$, mixed fcc/bcc phases resulting from a eutectic reaction were observed. Spinodal decomposition further occurred when the aluminum contents were higher than $x = 1.0$, and a single ordered bcc structure was obtained for aluminum contents larger than $x = 2.8$. The effect of high mixing entropy enhances the formation of simple solid-solution phases in the alloys with multiprincipal elements rather than intermetallic compounds and terminal solid solutions, and relaxes the constraints of Hume–Rothery rules. Nanosized phases formed and were attributable to the sluggish cooperative diffusion of many kinds of atoms to accomplish composition partition.

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